Holonomy-Aware Topological AlphaZero: Self-Play Go on Non-Orientable and Arbitrary-Genus Boards

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July 9, 2026

Abstract

Game-playing agents assume flat rectangular boards: the convolutional towers of AlphaZero and KataGo are defined over $\mathbb{Z}^2$ tensors and cannot even ingest a board on a sphere, a torus, or a Möbius band. We introduce *Geodesics*, a topological generalization of Go to arbitrary closed and bordered surfaces (spheres, tori, cylinders, Möbius bands, Klein bottles, real projective planes) and lattices in one and three dimensions, under any of several mesh families — and ask what architecture plays it optimally. We present a first implementation of a *Holonomy-Aware Topological AlphaZero* (HATZ): an AlphaZero-style self-play agent whose network (i) carries the reflection ($\mathbb{Z}_2$) part of the gauge structure that non-orientable surfaces provably require, via an orientation cocycle computed from the board's combinatorics and orthogonal (Cayley-parameterized) sheaf-style transport maps conditioned on it; (ii) is exactly equivariant to each board's automorphism group by construction, replacing the dihedral data augmentation that has no analogue off the square grid; (iii) pools through interlevel-set components of its *own predicted ownership field* — a learned Morse function whose partition is shaped by supervision rather than differentiable relaxation; (iv) laterally decomposes each position through an outcome-supervised edge filtration (a GIN-ε head predicting endpoint co-ownership) with per-node attention across filtration levels; and (v) reads the board as a cell complex, passing messages over vertex–edge–face incidences with curvature and localized-homology inputs and an ownership auxiliary head. The network is small (3.5×10^4 parameters), dependency-free, and every gradient is hand-derived and numerically verified. We verify the orientation cocycle against ground-truth orientability on eleven board types, demonstrate machine-precision equivariance, and report three pathologies specific to this setting — transposition-table ko cycles, a degenerate early-pass equilibrium under Tromp–Taylor scoring, and the failure of index-based tie-breaking (simulation of simplicity) to respect board symmetries when the Morse function is learned — together with their remedies. To our knowledge this is the first game-playing agent on non-orientable boards; a full transfer and ablation study is specified and in progress.

1 Introduction

The rules of Go never needed a flat grid. Stones occupy vertices of a graph; liberties are empty neighbors; a chain lives while its liberty set is non-empty; territory is graph reachability. Everything downstream of these definitions — capture, ko, life and death, Tromp–Taylor scoring —

*Draft v0.1. Code, boards, and the browser-playable implementation: https://samleventhal.com/subpages/geodesics/geodesics.html.

is intrinsically combinatorial. The 19×19 lattice is a historical accident of the plane. *Geodesics* takes the rules at their word and instantiates them on discrete surfaces: geodesic and Goldberg spheres, square and hexagonal tori, Möbius bands, Klein bottles, the real projective plane, cylinders, and cubic lattices in E^3, with vertex, edge, face, or full cell-complex incidence structures.

This generality breaks every assumption baked into modern game networks. AlphaZero [1] and KataGo [2] use residual convolutional towers over rectangular tensors; their 8-fold dihedral data augmentation exploits a symmetry group that simply does not exist on a sphere (whose geodesic boards have icosahedral symmetry) or a Klein bottle (whose isometries include orientation-reversing translations). More fundamentally, Weiler et al. [13] prove that convolution on a non-orientable surface cannot reduce its structure group to $SO(2)$: any consistent feature transport must be steerable under reflections, because parallel transport around an orientation-reversing loop returns frames mirrored. A network that ignores this is discontinuous across the orientation seam of a Möbius or Klein board.

We ask: *what is the right architecture for optimal play across domain geometries and topologies*, and we build a first honest draft of the answer. Our contributions:

1. **The first game-playing agent on non-orientable and arbitrary-genus boards.** We compute a representative orientation cocycle $\varepsilon \in Z^1(G; \mathbb{Z}_2)$ of the first Stiefel–Whitney class from the board's combinatorics alone (dual-BFS over face cycles when faces exist; the quotient-gluing rule for fundamental-domain grids), verify it against ground truth on eleven board types, and pass messages through ε-conditioned transport maps so that crossing the seam applies a genuinely different learned map (§4.1).

2. **Exact automorphism equivariance without augmentation.** Weight-shared message passing over the board graph commutes with every graph automorphism; we keep all inputs Aut-invariant by construction and verify equivariance to machine precision, including under the seam-preserving isometries of the Möbius band (§4.6). This replaces dataset-specific augmentation with intrinsic symmetry, with the strict generalization benefit of equivariance [19].

3. **Ownership-driven topological pooling.** The scalar field whose level structure defines the pooling regions is the network's own mid-depth ownership prediction — a *learned, supervised* Morse function, following the hierarchical-segmentation program of [20]. Regions are the connected components of its interlevel sets (Black-leaning, contested, White-leaning), a position-dependent, parameter-independent operator through which backpropagation stays exact, and — unlike basin partitions with simulation-of-simplicity tie-breaking — one that is exactly automorphism-equivariant (§4.5).

4. **Outcome-supervised filtration levels.** Following the homophily-filtration methodology of the companion manuscript [18] and graph filtration learning [16], a GIN-ε edge head predicts endpoint *co-ownership* — Go heterophily is literal: contact fights are heterophilous edges — and fixed thresholds over it produce nested edge-filtered levels (chain cores $\rightarrow$ contested full graph). Message passing runs per level with learned per-node attention across levels, so gradients reach the filter through supervision and attention while the masks themselves stay combinatorial (§4.3).

5. **A cell-complex, ownership-supervised value/policy net** with curvature (angle-defect) and localized-homology inputs, KataGo-style global pooling, and an ownership

head; 3.5×10^4 parameters, pure-`numpy` with hand-derived gradients verified numerically to 5×10^{-6} over every parameter (§4.4).

6. **Two self-play pathologies and their fixes**: PUCT descents that cycle forever through ko transpositions when the in-tree engine forgets history, cured by a positional-superko guard; and a degenerate early-pass equilibrium under Tromp–Taylor-plus-komi scoring that collapses the policy, cured by masking passes early in self-play (§5).

The literature intersection we occupy — game playing $\times$ gauge equivariance $\times$ non-orientable surfaces — is, to our knowledge, empty.

2 Related work

Game networks and size generality. KataGo trains one convolutional net across board sizes with masking and global-pooling bias layers and extrapolates beyond its training sizes [2]; Polygames obtains board-size invariance from fully convolutional trunks with global pooling [3]; ScalableAlphaZero replaces the CNN with a GIN and transfers from small to large Othello boards with an order-of-magnitude training-time reduction [4]. All operate on rectangular (or at least fixed-family) boards; none addresses topology.

GNN expressivity. Message passing is bounded by the 1-WL test [5, 6]; Go meshes are locally regular, which is precisely where 1-WL is weakest, motivating multi-aggregator updates [7], higher-order structure, and our cell-complex lifting, which is strictly more expressive than 1-WL [8].

Gauge equivariance. Gauge-equivariant CNNs on manifolds and meshes [11, 12] achieve equivariance to local frame choice via parallel transport; Weiler et al. [13] prove non-orientable surfaces force the structure group $O(2)$ (reflections included), demonstrating on a Möbius toy model. We implement the discrete reflection ($\mathbb{Z}_2$) part, which is the component non-orientability makes mandatory.

Topological deep learning. Simplicial and CW networks [8], combinatorial complex networks [9], and neural sheaf diffusion [10] pass messages over incidence structures with (in the sheaf case) learnable transport. Our boards *are* cell complexes, and our transport maps are a transfer-friendly sheaf: globally shared matrices conditioned on the edge's cocycle value, rather than per-edge free parameters.

Hierarchical pooling and topological segmentation. Graph U-Nets [14] and DiffPool [15] learn coarsenings; our pooling is the Morse–Smale multi-persistence hierarchy of a scalar field on the board, following the hierarchical GNN construction of [20], which gives a canonical segmentation into basins (territories) meeting along separatrices (contested frontiers).

3 Setting: Go on a surface

A *board* is a finite graph $G = (V, E)$, optionally equipped with a 2-cell structure (face cycles) making it a closed or bordered combinatorial surface, produced by `make_board(surface, mesh, resolution)`. The surfaces used here, with the smallest instances in our suite:

board	$\lvert V \rvert$	$\lvert E \rvert$	χ	orientable	seam edges
sphere (geodesic f=2)	42	120	2	yes	0
plane D^2 (5×5)	25	40	1	yes	0
cylinder (5×5)	25	45	0	yes	0
torus (5×5)	25	50	0	yes	0
Möbius (5×5)	25	45	0	no	5
Klein (5×5)	25	50	0	no	5
$\mathbb{RP}^2$ (5×5)	25	48	1	no	8
box E^3 (4^3)	64	144	—	—	0
3-torus (4^3)	64	192	—	—	0

Rules are graph-intrinsic (positional superko; Tromp–Taylor scoring with komi 7.5). Quotient boards are fundamental-domain grids with identifications; e.g. the Klein bottle identifies $(x+n_x, y) \sim (x, n_y-1-y)$ (a flip) and $(x, y+n_y) \sim (x, y)$; $\mathbb{RP}^2$ flips both pairs. The browser implementation and the training package construct adjacency independently and agree edge-for-edge (verified in tests), so trained agents transfer between them.

4 Architecture

4.1 Reflection gauge structure from combinatorics

A local orientation at a vertex is a $\mathbb{Z}_2$ gauge choice; an assignment $\varepsilon : E \to \{\pm 1\}$ records whether traversing an edge preserves or reverses it. Gauge transformations act by $\varepsilon'(u, v) = g(u)\,\varepsilon(u, v)\,g(v)$ for $g : V \to \{\pm 1\}$; the *holonomy* of a cycle, $\prod_{e\in\gamma} \varepsilon(e)$, is gauge-invariant, and the class $[\varepsilon] \in H^1(G; \mathbb{Z}_2)$ restricted to surface cycles represents the first Stiefel–Whitney class w_1: it is trivial iff the surface is orientable.

We compute a representative in two ways. When face cycles are available, a BFS over the dual graph assigns each face an orientation; a shared edge whose two incident faces traverse it incoherently under the assignment is a *seam* edge ($\varepsilon = -1$). For fundamental-domain grids the flipped gluing itself is the seam: x-wrap edges on Möbius and Klein boards, both wrap families on $\mathbb{RP}^2$.

Proposition 1. *The seam set produced by dual-BFS is a cocycle representing w_1; it is empty for some BFS iff the surface is orientable. On quotient grids, the flipped-gluing edge set is likewise a representative.*

Proof sketch. Dual-BFS fixes a gauge on faces along a spanning tree; conflicts can occur only on non-tree dual edges, and the conflict indicator is exactly the obstruction cocycle to extending the orientation, whose class is w_1. For quotient grids, cutting along the flipped gluing yields the orientable fundamental domain, so a loop reverses orientation iff it crosses the gluing an odd number of times. □

Empirically, the construction recovers ground-truth orientability on all eleven board types in our suite (including hexagonal-mesh Möbius bands and 3D lattices), with seam counts matching the gluing combinatorics (Table above).

4.2 ε-conditioned sheaf transport

Let $h_v \in \mathbb{R}^D$ be vertex features. Messages pass through restriction maps conditioned on the cocycle,

$$m_v \;=\; \tfrac{1}{|N(v)|}\sum_{u\in N(v)} \left(W_0 + \varepsilon(u,v)\, W_1\right) h_u \;=\; \big((Ah)\, W_0\big)_v \;+\; \big((A_\varepsilon h)\, W_1\big)_v, \qquad W_0, W_1 \in O(D), \tag{1}$$

with orthogonality enforced by the Cayley parameterization $W = (I-\Lambda)(I+\Lambda)^{-1}$, $\Lambda = T - T^\top$ for a raw parameter T (backward pass derived in closed form and verified numerically). Orthogonal restriction maps make the transport norm-preserving — genuine discrete parallel transport rather than arbitrary mixing — tightening the correspondence with connection sheaves. where A is the row-normalized adjacency and A_ε its ε-signed counterpart. Crossing the orientation seam therefore applies a different learned map — the mechanism by which neural sheaf diffusion [10] encodes orientation reversal and heterophily — but with W_0, W_1 *shared globally* rather than learned per edge, so the same parameters transfer across boards, and because ε enters linearly the layer reduces to two fixed aggregation matrices and remains trivially differentiable by hand.

Honest scope. This is the discrete reflection part of the $O(2)$ structure non-orientability mandates [13]; the continuous $SO(2)$ part (anisotropic kernels with rotational transport over the mesh's rotation system) is future work, and our aggregation is isotropic. Further, the network consumes ε in a fixed canonical gauge; holonomy-derived inputs are gauge-invariant, but the learned transport is gauge-covariant only up to reparameterization. We therefore train with *gauge augmentation*: each self-play game and each optimization step resamples g uniformly and uses $\varepsilon'(u,v) = g(u)\varepsilon(u,v)g(v)$, teaching the invariance the architecture does not yet guarantee.

4.3 Outcome-supervised filtration levels

Adjacent opposite-colored stones are heterophilous edges in the literal game-theoretic sense: contact fights concentrate heterophily, chains are homophilous cores. We port the homophily-based filtration methodology of [18] with class labels replaced by game outcomes. A GIN-ε head [5, 16] over the input embedding produces a symmetric per-edge filter $\phi(u,v) = \sigma(\mathrm{MLP}[g_u{+}g_v;\ |g_u{-}g_v|])$ supervised by *co-ownership*: whether the endpoints are owned by the same player under final Tromp–Taylor scoring (targets 1 / 0 / 0.5 for same/opposite/neutral). Fixed thresholds $(0.75, 0.5, \mathrm{full})$ give nested edge subsets — predicted chain cores through the contested full graph. The transport of Eq. 1 runs *per level* (the twisted aggregation matrices simply restrict to each kept edge set, so gauge structure and equivariance are untouched), and levels combine by per-node attention $h_v = \sum_\ell \alpha_{v,\ell}\, m_v^{(\ell)}$, $\alpha_v = \mathrm{softmax}_\ell(m_v^{(\ell)} \cdot a_\ell)$, generalizing the filtration attention of [18]. Gradients reach the filter through its supervision and through the attention weights — always differentiable — while the level masks themselves are combinatorial structure: supervised, not relaxed, which sidesteps the piecewise-constant-membership problem entirely.

4.4 Cell ranks and inputs

The board is read as a complex: vertex features always; edge features always (seam flag, endpoint color pattern); face features when face cycles exist. Per layer ℓ,

$$H_V \leftarrow \mathrm{relu}\big(H_V W_s + (AH_V)W_0 + (A_\varepsilon H_V)W_1 + (M_{EV}H_E)W_{ev} + \bar{h}\, W_g + b\big), \tag{2}$$

$$H_E \leftarrow \mathrm{relu}\big(H_E W_{es} + (M_{VE}H_V)W_{ve} + (M_{FE}H_F)W_{fe} + b_e\big), \qquad H_F \leftarrow \mathrm{relu}(\cdots), \tag{3}$$

with $M_{\cdot\cdot}$ fixed incidence-mean operators and $\bar{h}$ the global mean (KataGo-style pooling bias [2]). Inputs are Aut-invariant by construction: stone/empty indicators from the mover's perspective, normalized degree, discrete curvature (the Gauss–Bonnet angle defect $2\pi - \sum_i \theta_i$, whose total is $2\pi\chi$), localized H_0 data (own/opponent group size and liberty counts — the persistent-homology features in their cheapest useful form), and global scalars $[\,\mathbf{1}[w_1 \neq 0],\ \chi/|V|\,]$. Heads: a per-vertex policy logit plus a pooled pass logit; a pooled tanh value; and a per-vertex tanh *ownership* head supervised by final Tromp–Taylor territory, KataGo's auxiliary target adapted to graphs. The ownership head is applied twice with shared weights: at mid-depth (where its prediction becomes the pooling Morse function) and at the output, both supervised. Total: 38,341 parameters at width 32×3 layers — $\sim$150 KB, comfortably client-side. All gradients are hand-derived in `numpy`, including the Cayley inverse, the per-node attention softmax, and the filter head; central-difference verification over every parameter (with position structure frozen, since it is deliberately stop-grad) gives worst relative error 4.3×10^{-7} (Möbius) and 6.0×10^{-6} (tetrahedron), and $\|W^\top W - I\|_\infty = 6.7 \times 10^{-16}$ for the transported maps.

4.5 Ownership-driven topological pooling

The coarsening rung uses a *learned* Morse function: the mid-depth ownership prediction $\hat{o}$. Regions are the connected components of its interlevel sets $\{\hat{o} \geq t\}$, $\{|\hat{o}| < t\}$, $\{\hat{o} \leq -t\}$ at the median-$|\hat{o}|$ threshold — predicted Black territory, contested frontier, predicted White territory, the sublevel-set objects native to persistence. Features mean-pool to regions, message-pass once on the region graph, and unpool residually. The partition depends on the position and on $\hat{o}$ but is treated as stop-grad structure; the field is shaped by its own supervision ("supervised, not relaxed"), so backpropagation through the block is exact.

We initially used the ascending/descending *basin* partition of [20], and our equivariance tests caught a failure worth recording: with a learned field, symmetric positions produce value *plateaus*, and the index-based tie-breaking (simulation of simplicity) that discrete Morse theory uses for genericity then splits them arbitrarily — on the empty geodesic sphere the twelve equal pentavalent maxima carved the board into twelve index-dependent basins, and on the Möbius band two *chiral twin* seam-corner plateaus, exchanged by the band's reflection, merged through different union-find chains. Simultaneous merging of equal-persistence families repairs the first case but not the second. Interlevel-set components repair the class: components map to components under any symmetry of the field, so the partition is exactly automorphism-equivariant, verified to machine precision. Basin/separatrix structure remains in the fixed built-in engine, and a differentiable soft-ascent basin variant is a natural extension.

4.6 Equivariance

Proposition 2. *If every input feature is invariant under* $\mathrm{Aut}(G)$ *(and, where used, under automorphisms of the complex and of ε), the network is exactly* Aut-*equivariant: for any automorphism σ, $\pi(\sigma \cdot s) = \sigma \cdot \pi(s)$ and $v(\sigma \cdot s) = v(s)$.*

Proof sketch. Each layer is a composition of permutation-equivariant operators ($A, A_\varepsilon, M_{\cdot\cdot}, S, P$ conjugate by the permutation; pointwise nonlinearities and global means commute with it) with shared weights. □

Verified to machine precision for the full v2 stack (filtration levels, attention, Cayley transport, ownership-driven pooling): the empty-board policy is constant on the WL orbits of the geodesic sphere and the torus, and exactly symmetric under the seam-preserving Möbius reflection — all spreads at floating-point noise. The filter ϕ is symmetric and permutation-equivariant by construction, so level masks map to level masks under automorphisms. On non-orientable boards the *gauge-dependent* remainder — seam placement — is the component gauge augmentation trains away; we report this rather than hide it.

5 Self-play training, and two pathologies

Training is standard AlphaZero: PUCT (c=1.5, Dirichlet root noise), visit-count policy targets with temperature 1 for the first ten moves, replay buffer, Adam. The loss is $\mathrm{CE}(\pi, p) + (z - v)^2 + \frac{1}{2}\,\mathrm{MSE}(\hat{o}_{\text{final}}) + \frac{1}{4}\,\mathrm{MSE}(\hat{o}_{\text{mid}}) + \frac{1}{4}\,\mathrm{BCE}(\phi)$, all targets derived from final Tromp–Taylor territory. Gauge augmentation resamples the cocycle per game and per optimization step. Two failures specific to this setting are worth recording.

Transposition ko cycles. Our search merges states in a transposition table keyed by (stones, mover, passes) and re-seats a scratch engine per step, which forgets ko history. Once two ko-mirror nodes exist with PUCT argmaxes pointing at each other, a descent cycles $A \to B \to A$ forever — the process hangs, typically from the second iteration once the tree is deep enough to contain a ko. The fix is the search-side analogue of the game's superko rule: any state repeated within a single descent scores as a draw and terminates (with a depth cap as backstop). Three hundred simulations through a live ko now complete in bounded time; the guard is a permanent regression test.

The early-pass equilibrium. Under Tromp–Taylor scoring with positive komi, an immediate double pass is a *guaranteed White win.* Shallow search discovers this instantly: White's visits collapse onto pass, half the replay buffer becomes one-hot pass targets, and the learned policy passes as both colors — losing every evaluation game to a random mover. (Loss decreases throughout; the curve looks healthy while the policy degenerates.) KataGo guards its self-play against early passing for the same reason [2]; we mask the pass action from search targets for the first $|V|/2$ moves, after which games run full length and evaluation strength reflects play rather than the scoring artifact.

6 Preliminary verification and evaluation protocol

What is verified now. Cocycle-vs-orientability on 11 board types; full-parameter gradient checks (4.5×10^{-6}, 5.8×10^{-7}); machine-precision equivariance as above; Gauss–Bonnet consistency of the curvature inputs (tetrahedron defect = π per vertex); browser/Python adjacency parity edge-for-edge on cylinder, Klein, and $\mathbb{RP}^2$ boards; end-to-end training and serving (checkpoint $\to$ WebSocket bridge $\to$ browser opponent, including seam reconstruction on remote boards). Short v2 training runs on the non-orientable trio (Möbius, Klein, $\mathbb{RP}^2$; 24 simulations, six iterations) show healthy optimization of the five-term loss after the pass guard ($4.20 \to 3.39$; parity with a random opponent by iteration four) — reported as smoke evidence, not results.

Protocol (in progress). (i) *Zero/few-shot transfer*: train on k topologies, evaluate Elo on held-out ones (e.g. train sphere+torus, test Klein), against fixed anchors (random; the hand-designed Morse–Smale engine; a Stage-1 GNN of matched size). (ii) *Ablations*: remove in turn the ε-transport (W_1=0), the MSC block, the ownership head, curvature inputs, and gauge augmentation; measure Elo and gauge-sensitivity. (iii) *Ceiling*: on the plane boards only, compare against KataGo as an upper reference. (iv) *Symmetry*: measure sample efficiency versus an augmentation-based baseline, testing the equivariance dividend of [19] in self-play.

7 Limitations and future work

No continuous $SO(2)$ steerability yet (isotropic aggregation over the rotation system); gauge invariance is learned, not guaranteed — a strictly equivariant formulation via the orientation double cover is the natural next step; filtration thresholds are fixed hyperparameters rather than learned quantiles; the soft-ascent basin variant that would restore separatrix structure differentiably remains future work; persistent-homology inputs beyond interlevel components (H_1 for eyes/ko) are pending; automorphism groups are used for verification, not yet target symmetrization; and the empirical study above is the actual test of the thesis. A browser export of the network is mechanical — the Morse–Smale machinery already ships in the client-side engine — and would make every trained agent playable with no installation.

8 Conclusion

Go's rules are indifferent to topology; the field's architectures are not. We built the first agent that plays where the boards get strange — carrying exactly the reflection structure that non-orientability forces, the automorphism symmetry each board actually has, and a topological pooling hierarchy with guarantees — small enough to ship in a browser tab, honest about which invariances are proved, learned, and pending.

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